

September Crossword

Kevin Chen, Wenhao Lu, Aaron Qin

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ONE DOWN

A scarecrow is walking across a cornfield grid from the bottom-left corner $(0, 0)$ to the top-right corner $(7, 7)$. On each step, the scarecrow can move one unit in the positive x direction or one unit in the positive y direction. In the middle of the field is a pumpkin patch, with corners at $(2, 2)$, $(5, 2)$, $(5, 5)$, and $(2, 5)$, which the scarecrow must *avoid entirely*. Given that the number of different paths the can scarecrow take to reach $(7, 7)$ without ever stepping inside the pumpkin patch is n , find the sum of the exponents and distinct prime factors in the prime factorization of n .

THREE ACROSS

One day in the town of Mexville, a strong gust of wind blows exactly 10 leaves off a tree. The next day, a stronger gust blows off exactly 1010. In general, on day n , the leaves appear as n repetitions of “10” (e.g., 101010 when $n = 3$).

Believing the winds have supernatural powers, the townspeople decide to measure each day’s arrangement by taking the number of leaves blown off and dividing it by 101^n . The townspeople keep track by adding the value on days when the wind blows north and subtracting the value on days when it blows south, alternating direction each day.

Thus the townspeople write the infinite series:

$$S = \frac{10}{101} - \frac{1010}{101^2} + \frac{101010}{101^3} - \frac{10101010}{101^4} + \dots$$

Given that S converges to a simplified fraction $\frac{m}{n}$, find the sum of digits of $m + n$.

ONE ACROSS

At a Thanksgiving table, 9 chairs are placed in a circle. There are 4 adults and 5 children, of which 3 are boys and 2 are girls. However, the 3 boys are troublemakers and cannot sit next to one another. The number of possible seating arrangements is $100m$. Find $\sqrt{m}$.

TWO DOWN

In the 21st century (the years 2000-2099), how many Septembers have exactly five Saturdays?